



5.6 Volumes of revolution (HL)

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

When the region under a curve $y=f(x)$ between $x=a$ and $x=b$ is rotated 360° about the x -axis, the resulting solid has volume $V = \pi \int [a \text{ to } b] y^2 dx$. Similarly, rotating a region between $x=g(y)$ and the y -axis about the y -axis gives $V = \pi \int [c \text{ to } d] x^2 dy$. These formulas come from summing infinitesimally thin circular discs of radius y (or x) and thickness dx (or dy).

Worked example: follow the reasoning

Find the volume generated when the region under $y=\sqrt{x}$, between $x=0$ and $x=4$, is rotated 360° about the x -axis.

$$V = \pi \int [0 \text{ to } 4] (\sqrt{x})^2 dx = \pi \int [0 \text{ to } 4] x dx = \pi [x^2/2] \text{ from } 0 \text{ to } 4 = \pi(16/2 - 0) = 8\pi.$$

Quick check · multiple choice

The region under $y=x$, from $x=0$ to $x=3$, is rotated about the x -axis. What is the volume, in terms of π ?

- A 3π
- B 9π
- C 18π
- D 27π

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The curve $y = x^2 + 1$ for $0 \leq x \leq 2$ is rotated 360° about the x -axis to form a solid. [6]
- (a) Write down the integral for the volume of the solid.
- (b) Evaluate this integral to find the exact volume, in terms of π .

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $V = \pi \int_{[0 \text{ to } 2]} (x^2+1)^2 dx.$

(b) $(x^2+1)^2 = x^4+2x^2+1.$ $\int (x^4+2x^2+1)dx = x^5/5 + 2x^3/3 + x.$

At $x=2$: $32/5 + 16/3 + 2.$ At $x=0$: $0.$

$32/5+16/3+2 = (96+80+30)/15 = 206/15.$

$V = \pi(206/15) = 206\pi/15.$

Quick check answer

B. $V=\pi \int_{[0 \text{ to } 3]} x^2 dx = \pi[x^3/3] \text{ from } 0 \text{ to } 3 = \pi(9-0) = 9\pi.$

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.